

Architectural Evolution of Latency-Critical Inference Systems for Edge and vRAN Environments

The Cellular Edge Paradigm and the Coordination Baseline

The proliferation of deep neural network (DNN) applications across mobile, autonomous, and Internet of Things (IoT) ecosystems has forced a structural migration of computational workloads from centralized cloud datacenters to the network edge. This architectural shift, often realized through Multi-Access Edge Computing (MEC) and virtualized Radio Access Network (vRAN) deployments, is necessitated by the constrained computational capabilities and battery limitations of end-user devices. These devices are fundamentally incapable of executing massive inference workloads locally while meeting the strict latency service-level objectives (SLOs) demanded by advanced artificial intelligence applications. However, the deployment of latency-critical inference serving over shared 5G and 6G cellular networks introduces profound infrastructural challenges. In these environments, concurrent workloads continuously contend for highly constrained resources across three interconnected stages: the radio uplink, the edge computational matrix, and the radio downlink.¹

The seminal work on the CORA system, presented at ACM CoNEXT 2025, establishes a critical and transformative baseline for addressing these end-to-end cyber-physical challenges. Historically, approaches to edge inference have primarily focused on uplink-centric designs tailored to specific tasks, such as rudimentary video analytics, thereby lacking the versatility and systematic resource orchestration required for a heterogeneous mix of modern DNN workloads.² CORA dismantles this siloed approach by operating as an end-to-end coordination mechanism between the RAN and the edge server. CORA systematically negotiates and dynamically adjusts the latency budgets across the networking and compute stages.² By analyzing the unique architectural characteristics and latency requirements of each specific DNN task, CORA intelligently balances resource demands across the radio frequency spectrum and the edge compute domains to mitigate stage-specific contention.²

A defining feature of the CORA architecture is its ability to align the underlying resource schedulers—such as the MAC layer scheduler allocating Physical Resource Blocks (PRBs) in the RAN and the GPU scheduler managing execution in the edge server—with these dynamically generated per-stage budgets.³ This enables seamless end-to-end coordination without requiring any architectural modifications to the end-user host devices, ensuring broad backward compatibility.³ Empirical evaluations conducted on an over-the-air testbed demonstrate that CORA achieves transformative performance gains; it serves 3.2× more

requests within the targeted latency budget and reduces the 95th percentile tail latency by a factor of 2.1× compared to uncoordinated, baseline systems.³

Building upon the orchestration foundation laid by CORA, it is imperative to deeply analyze the specific inference workloads that populate these cellular edge networks and the state-of-the-art (SOTA) inference systems engineered to execute them. Between 2022 and 2026, research in top-tier systems and machine learning venues (e.g., OSDI, NSDI, MLSys, SIGCOMM, ATC) has produced highly specialized, hardware-aware architectures designed for Large Language Models (LLMs), high-concurrency video analytics, and 3D LiDAR spatial perception. The subsequent analysis deconstructs these SOTA systems, extracting their core architectural mechanisms regarding memory and cache management, request scheduling and batching, session and state management, and hardware parallelization, while rigorously evaluating their viability for deployment in resource-constrained edge and vRAN environments.

Identification of Inference Workloads Evaluated in CORA

To understand the systems required for edge inference, one must first isolate the specific workloads evaluated within the CORA framework. The CORA system is explicitly designed to handle "diverse DNN tasks" that contend for resources across the uplink, computation, and downlink stages.¹ Based on the evaluation parameters and related research surrounding the CORA deployment, these latency-critical workloads are categorized into several distinct neural network paradigms, each exhibiting unique stress profiles on the network and compute infrastructure.²

The first primary workload category is Computer Vision and Object Detection. Models such as YOLO (You Only Look Once), ResNet (Residual Networks), and MobileNet represent the standard proxies for visual inference tasks.⁶ These workloads are highly uplink-intensive, as they require the transmission of high-resolution image frames from the mobile device to the edge server. Computationally, they rely heavily on dense matrix multiplications and convolutional operations. ResNet and YOLO architectures aggressively consume GPU Streaming Multiprocessors (SMs) but have relatively bounded and predictable memory access patterns once the weights are loaded into VRAM.⁶ MobileNet, specifically engineered with depthwise separable convolutions, is designed to minimize parameter count, making it highly viable for environments where edge compute must be minimized, allowing CORA to allocate more of the latency budget to the unpredictable radio uplink.⁶

The second major workload category is Video Analytics. Unlike static image classification, video analytics involves continuous, high-bandwidth streams.² This workload generates immense and sustained pressure on the cellular uplink. At the compute stage, it introduces the necessity of hardware-accelerated video decoding (e.g., H.264/H.265) before neural network inference can even begin. The temporal nature of video analytics means that the system must maintain spatial-temporal session state across frames, creating complex memory management

requirements to track objects over time without exhausting the edge server's cache.¹¹

The third workload category encompasses Natural Language Processing (NLP) and Large Language Models (LLMs), represented in evaluation baselines by models like BERT (Bidirectional Encoder Representations from Transformers) and its autoregressive successors.⁶ NLP workloads present an inverted resource profile compared to computer vision. The uplink and downlink requirements are minimal, consisting merely of text tokens. However, the compute stage is extraordinarily constrained by memory capacity and memory bandwidth.¹⁵ The autoregressive nature of LLM generation requires the system to maintain a massive, continuously expanding Key-Value (KV) cache for every active session, making memory management the primary architectural bottleneck.¹⁵

By validating CORA against these diverse DNN tasks, the system proves its capability to orchestrate environments where bandwidth-heavy workloads (video/vision) and memory-heavy workloads (NLP) concurrently contend for the same shared edge infrastructure.¹

State-of-the-Art Inference Architectures Summary Matrix

The following matrix synthesizes the premier inference systems published between 2022 and 2026. It categorizes them by target workload, publication venue, and their defining architectural mechanisms, establishing a comparative framework for their deployment in vRAN environments.

System Name	Target Workload	Venue / Year	Memory & Cache Mechanisms	Scheduling & Session Mechanisms	Hardware Parallelization	Edge/vRAN Viability
vLLM ¹⁵	LLMs / NLP	SOSP 2023	PagedAttention; OS-level virtual memory paging for KV cache.	Continuous batching; dynamic sequence length management.	High-throughput kernel fusion via xFormers.	High; maximizes limited edge GPU memory by eliminating internal fragment

						ation.
Sarathi-Serve ¹⁶	LLMs / NLP	OSDI 2024	Compute /memory bound state separation.	Chunked prefilling; iteration-level scheduling.	Overlaps ALU saturation with memory bandwidth saturation.	High; smooths latency spikes in shared environments by preventing head-of-line blocking.
WEAVER ²¹	LLMs / NLP	ATC 2025	Zero-copy shared memory QKV polling; attention offloading.	Pipeline multi-model state orchestration.	GPU-driven dynamic control flow bypassing CPU intervention.	Very High; enables multi-tenant edge nodes to serve hundreds of cold models without memory bloat.
PacketGame ¹¹	Video Analytics	SIGCOMM 2023	Bypasses decode memory allocation via packet drops.	Multi-stream packet gating; combinatorial optimizer ; temporal state.	Hardware decoder ASIC offloading via ingress traffic gating.	Critical; mitigates uplink and decode bottlenecks before AI inference , perfect for cellular

						integration.
Dynamic SinkCache ²⁴	Video LLMs	arXiv 2025	Task-focused hybrid memory retention.	Sliding window context management; spatial-temporal session tracking.	Bounded KV cache allocation for long-context streams.	Moderate ; requires precise tuning to avoid dropping critical spatial-temporal context in constrained nodes.
FALO ⁹	3D LiDAR	CVPR 2025	1D sequence mapping from sparse 3D voxels.	Deterministic spatial array scheduling.	Eliminates irregular memory accesses; maximizes hardware prefetching.	Very High; specifically designed to execute autonomous driving perception on resource-constrained edge hardware.
LiFT ²⁹	3D LiDAR	arXiv 2025	Minimal block RAM usage via 2D cells.	Static spatial state management.	FPGA-specific INT8 quantization; Dual-bound Pillar	High; enables ultra-low latency execution (20.73 GMACs)

					Feature Net.	directly on edge telecommunication hardware.
SLA-Driven Framework ³⁰	Heterogeneous	MLSys 2022	Proxy-based dynamic metric tracking.	Heuristic per-stage SLAs; dynamic auto-scaling.	Routes across heterogeneous accelerators based on task requirements.	Moderate to High; provides cluster-level abstraction applicable to centralized RAN (C-RAN) datacenters.
μ-Serve ₄	Heterogeneous	2023	Power-constrained thermal throttling memory bounds.	Deadline-aware latency slack monitoring.	Power-aware dynamic GPU frequency and voltage scaling.	Essential; addresses severe power and thermal constraints present in remote edge cell towers.

Large Language Models and Natural Language Processing Architectures

The deployment of Large Language Models (LLMs) at the cellular edge represents a profound paradigm shift in mobile computing. End-user applications increasingly demand real-time

natural language processing, semantic search, and generative capabilities that vastly exceed the memory and compute limits of mobile processors. However, executing LLMs on edge servers introduces severe memory bottlenecks, primarily due to the autoregressive nature of token generation and the massive footprint of the Key-Value (KV) cache. Between 2023 and 2026, system architectures have radically evolved to address these specific computational and memory boundaries.

Memory and Cache Management Mechanisms

The most fundamental breakthrough in LLM inference memory management was the introduction of PagedAttention, primarily popularized by the vLLM system presented at SOSP 2023.¹⁵ In traditional autoregressive generation, the KV cache for each active sequence must be stored in contiguous GPU memory. Because sequence lengths are highly variable and unpredictable a priori, traditional systems allocate memory based on maximum possible sequence lengths. This static allocation results in catastrophic internal memory fragmentation (allocated but unused memory) and external fragmentation (free memory scattered in blocks too small to use).¹⁵

PagedAttention addresses this by conceptualizing the GPU's KV cache as an operating system's virtual memory space. Memory is divided into fixed-size blocks (pages), allowing the KV cache of a single sequence to be entirely non-contiguous in physical memory.¹⁵ The system maintains a block table that maps logical tokens to physical GPU memory pages. This architecture achieves near-zero waste in KV cache memory. By eliminating fragmentation, the system can batch significantly more requests simultaneously, radically increasing the overall throughput of the edge server.¹⁵

Building upon the necessity of memory optimization in multi-tenant edge environments, the WEAVER system, introduced at USENIX ATC 2025, tackles the specific problem of serving tens to hundreds of different LLM models concurrently.¹⁹ In typical edge serving platforms managed by telecommunication operators, a few "hot" models receive the vast majority of requests, while numerous "cold" models (e.g., highly specialized fine-tuned models) consume GPU memory without providing high utilization.²² Dedicating hardware instances to cold models leads to severe underutilization, while dynamically multiplexing different models with standard model parallelism introduces prohibitive communication overheads over the PCIe bus.²²

WEAVER proposes "workload weaving," an architectural mechanism that offloads the highly memory-intensive attention operators of hot models directly to the execution pipelines of running cold models.²¹ This mechanism effectively treats the entire edge cluster's memory pool as a unified resource for attention computation. By utilizing the idle memory bandwidth of the cold models to process the attention states of the hot models, WEAVER achieves high GPU memory utilization while maintaining exceptionally low communication costs.²¹

Request Scheduling and Batching Architectures

Continuous batching has become the indispensable scheduling mechanism for LLM inference,

entirely replacing static batching. Under static batching, because different requests in a batch finish at different times, the GPU is forced to idle while waiting for the longest sequence in the batch to complete generation.¹⁸ Continuous batching dynamically manages the execution state, allowing the system to inject new queries into the execution batch immediately as other queries finish.¹⁵

However, continuous batching alone does not resolve the fundamental tension between the prefill phase and the decoding phase in LLM inference. The prefill phase (processing the input prompt) is highly compute-bound, utilizing massive, dense matrix multiplications to process hundreds of tokens simultaneously. Conversely, the decoding phase (generating output tokens one by one) is heavily memory bandwidth-bound, as the entire model weight matrix and KV cache must be loaded from High Bandwidth Memory (HBM) to the SMs for every single token generated. When a new request enters a continuous batch, its massive prefill computation monopolizes the GPU's SMs, causing a severe latency spike for all other requests currently in the decoding phase. This phenomenon, known as head-of-line blocking, violates the strict latency SLOs required by systems like CORA.²

To tame this throughput-latency tradeoff, the Sarathi-Serve architecture, presented at OSDI 2024, introduces chunked prefilling combined with iteration-level scheduling.¹⁶ Sarathi-Serve fractures the monolithic, latency-inducing prefill computation into smaller, manageable chunks (e.g., 256 or 512 tokens).¹⁶ During iteration-level scheduling, the system issues a single iteration to the GPU that contains both a chunk of prefill tokens from a new request and a set of decode tokens from ongoing requests.¹⁶ This architectural mechanism achieves a perfect hardware equilibrium: the compute-heavy prefill chunk saturates the GPU's arithmetic logic units (ALUs), while the decode tokens saturate the memory bandwidth.¹⁶ By structurally aligning the computational density of each iteration, Sarathi-Serve drastically reduces tail latency for token generation, ensuring that latency-critical tasks on the edge do not suffer from unpredictable jitter, making it highly synergistic with CORA's latency budgeting.⁴

Session Management and Hardware Parallelization

To implement aggressive offloading techniques without bottlenecking the system through CPU interventions, systems have adopted advanced hardware parallelization and state synchronization strategies. WEAVER mitigates the blocking caused by offloading attention operators to cold models through a novel GPU-driven dynamic control flow mechanism.²¹ Traditional distributed offloading requires the CPU to orchestrate data transfers, memory allocations, and kernel launches, which incurs unacceptable microsecond-to-millisecond latency penalties over the PCIe bus.

In WEAVER's highly parallel pipeline, the sender (the hot model executing its feed-forward layers) writes the Query, Key, and Value (QKV) matrices directly into GPU shared memory and updates a task counter atomically.²¹ The receiver (the cold model) operates a continuous polling kernel entirely on the GPU to select the task, executes the attention mechanism, writes the output back to shared memory, and updates a completion counter.²¹ The sender purely waits

for this output on the device. This CPU-bypass architecture allows for microsecond-level synchronization, turning independent parallel GPUs in an edge node into a tightly coupled accelerator fabric capable of managing complex session states autonomously.²¹

Furthermore, for environments requiring extreme scale and predictability, architectures like Chronos (OSDI 2025) explore prescheduled circuit switching for LLM fabrics.³⁶ Unlike packet switching, which handles unpredictable traffic, Chronos leverages the highly predictable nature of transformer architectures. It analyzes the inference code a priori and generates a sequence of permutations that preschedule a hardware crossbar switch.³⁶ For complex, dynamic states like Mixture-of-Experts (MoE) routing, Chronos utilizes Birkhoff-von Neumann decomposition in nanoseconds to schedule circuits on demand, drastically reducing power consumption and increasing switching capacity in densely packed edge clusters.³⁶

Viability for Edge and vRAN Environments

The viability of these LLM architectures for edge/vRAN environments is exceptionally high, provided they are integrated with RAN-aware coordinators like CORA. Edge servers located directly at cellular base stations (e.g., at the Distributed Unit or Centralized Unit in a vRAN split) suffer from strict space, power, and thermal limitations. They cannot host dozens of top-tier, datacenter-grade GPUs. Therefore, maximizing the utility of a single GPU is paramount.

The PagedAttention mechanism ensures that the limited VRAM is not wasted on fragmentation, allowing an edge node to support more concurrent users per cell sector.¹⁵ Sarathi-Serve's chunked prefilling is perhaps the most critical for vRAN viability; cellular networks guarantee Quality of Service (QoS) through strict, millisecond-level latency bounds. Eliminating the prefill-induced jitter ensures the edge server rigorously respects the compute latency budgets assigned by the CORA coordinator.⁴ Furthermore, WEAVER's workload weaving allows telecom operators to offer hundreds of specialized fine-tuned models to enterprise edge customers (e.g., specialized healthcare or legal LLMs) without requiring dedicated hardware for each tenant, ensuring the economic viability of multi-tenant edge deployments.²¹

High-Concurrency Video Analytics and Spatial-Temporal Architectures

Video analytics constitutes one of the most resource-intensive workloads over cellular networks, heavily utilizing both the radio uplink for camera streaming and the edge server for spatial-temporal decoding and AI inference. Smart city infrastructure, autonomous drones, and augmented reality (AR) headsets generate massive, continuous video streams that must be processed in real-time to be of any operational value.¹¹

Hardware Parallelization and Decoding Mechanisms

The primary bottleneck in end-to-end video inference is frequently not the execution of the neural network itself, but the preceding stages of network transmission and hardware video

decoding. The PacketGame architecture, introduced at ACM SIGCOMM 2023, provides a revolutionary hardware-aware mechanism for multi-stream packet gating to enable concurrent video inference at massive scale.¹¹ Traditional frame filtering systems execute after the video has been fully decoded. This paradigm wastes massive amounts of power, SM cycles, and memory bandwidth on decoding and transferring frames that the system will ultimately discard.¹¹

PacketGame implements a highly efficient architectural module situated logically between the network packet parser and the hardware video decoder.¹¹ Instead of decoding all incoming network packets, it utilizes a packet gating selector that filters data at the ingress network level based on specific packet metadata, such as the video codec, picture type (I-frames vs. P-frames), and the payload packet size.¹¹ This pre-decode gating drastically alleviates the computational pressure on the edge server's hardware decoding ASICs (Application-Specific Integrated Circuits) and the subsequent deep learning inference pipelines. By intelligently dropping non-essential packets, PacketGame improves system concurrency by up to 4.8× and saves up to 79.3% in decoding computational costs, effectively solving the ingestion bottleneck.¹¹

Request Scheduling and Contextual Session State

To ensure that aggressive packet gating does not destroy the semantic integrity of the video stream or irreparably disrupt the session state of the spatial-temporal inference model, PacketGame relies on mathematically rigorous scheduling and state estimation mechanisms. It deploys a sliding window-based Temporal Estimator that leverages online feedback and historical gating decisions to estimate the statistical probability that an incoming packet contains redundant visual information.¹¹

Furthermore, PacketGame utilizes a Contextual Predictor, structured as a lightweight multi-view neural network, to generate a deterministic gating confidence score.¹¹ This network processes three distinct views of the incoming state: the packet sizes of independent I-frames, the sizes of temporally dependent P-frames, and the probability output from the Temporal Estimator.¹¹ By fusing these distinct inputs through dense layers, the system maintains a robust contextual understanding of the video session's state without actually analyzing the decoded pixel data.

The ultimate scheduling of these streams is governed by a Combinatorial Optimizer that manages cross-stream coordination under a strict, hardware-defined decoding budget (B).¹¹ It greedily prioritizes and selects packets based on a computed ratio of the gating confidence to the decoding cost, while strictly accounting for the complex dependency structures within a standard Group of Pictures (GOP).¹¹ If an essential packet is selected for decoding, the optimizer algorithmically guarantees that all requisite reference packets are also scheduled for decoding. The architecture achieves a proven theoretical $1 - c/B$ approximation ratio

(where c is the maximal decoding cost of a single packet) and maintains an $O(\sqrt{T})$ regret bound based on multi-armed bandit theory.¹¹

Memory Management in Video LLMs

As edge video analytics transitions from standard CNN-based object detection (e.g., YOLO) to multimodal Video Large Language Models (Video LLMs), memory management becomes vastly more complex. Architectural evaluations of memory mechanisms for long-context video comprehension published in 2024 and 2025 demonstrate that standard caching strategies utterly fail when processing continuous video.²⁴ Relying solely on a sliding window of recent context or a static window of initial context results in catastrophic model degradation, as the multimodal model either forgets long-term spatial objectives or completely misses real-time state changes occurring in the stream.²⁶

The SOTA solution to this memory crisis is the Dynamic SinkCache mechanism.²⁴ This hybrid, task-focused memory system selectively retains critical spatial-temporal context within the KV cache while aggressively evicting redundant or low-information frames. It has been empirically proven to outperform both unbounded KV caches (which quickly trigger Out-Of-Memory errors on edge nodes) and standard static sinks.²⁶ Dynamic SinkCache ensures that the model's long-term memory allocation is persistently and exclusively focused on the primary analytical goal, preventing the session state from being diluted by irrelevant background changes or temporal noise.²⁵

Viability for Edge and vRAN Environments

The PacketGame architecture and Dynamic SinkCache mechanisms are quintessential for cellular edge viability. In a vRAN architecture governed by orchestration systems like CORA, the cellular uplink is the most severely constrained and contested resource.¹ Dozens of high-definition video cameras must constantly contend for Physical Resource Blocks (PRBs) on the 5G uplink.

A profound architectural integration occurs when the Contextual Predictor in PacketGame feeds its confidence scores directly back into the MAC layer scheduler of the vRAN. If the edge server determines that an upcoming P-frame is highly likely to be redundant based on the Temporal Estimator, the vRAN can dynamically revoke the uplink grant for that specific camera device, immediately reallocating the radio spectrum to other latency-critical tasks.⁴ This synthesis of application-layer gating and MAC-layer scheduling perfectly embodies the end-to-end coordination that CORA advocates, creating a unified, cyber-physical optimization loop that extends from the lens of the camera to the SMs of the edge GPU.² Frameworks like EdgeEye further enhance this viability by providing high-level, task-specific APIs that allow developers to transform complex models into deployable, edge-optimized components with minimal friction, wrapping these advanced mechanisms in accessible interfaces.⁴⁰

3D LiDAR and Autonomous Perception Architectures

Autonomous driving, advanced driver-assistance systems (ADAS), and robotic navigation depend heavily on 3D LiDAR point clouds to construct highly precise representations of the physical environment. Unlike 2D images, LiDAR data is inherently unstructured, unordered, and highly sparse.⁴³ The inference systems processing this data face unique, severe architectural challenges, particularly when deployed on resource-constrained vehicle-mounted edge devices or MEC nodes situated at roadside units (RSUs) in Vehicle-to-Infrastructure (V2I) networks.⁴⁴

Hardware Parallelization and Data Structures

Traditional LiDAR 3D object detection systems rely on sparse convolutions or massive, multi-head transformer architectures. While highly effective on datacenter-grade hardware with virtually unlimited power budgets, these algorithms map poorly to edge devices due to their highly irregular memory access patterns.²⁷ Sparse convolutions require complex hash tables and extensive pointer chasing to locate valid data points in 3D space, which aggressively thrashes the L1/L2 caches and severely stalls the parallel execution pipelines of the GPU.²⁷

The FALO (Fast and Accurate LiDAR 3D Object Detection) architecture, introduced in 2025, provides a highly hardware-friendly approach designed explicitly for resource-constrained edge devices.²⁷ FALO circumvents the memory access bottlenecks by fundamentally transforming the spatial data structure. After standard voxelization of the 3D point cloud, FALO algorithmically arranges the sparse 3D voxels into a contiguous 1D sequence.⁹ This architectural translation converts scattered, indirect memory reads into linear, highly predictable memory accesses.²⁷ By mapping the computation to linear arrays, FALO maximizes the hardware cache hit rate and ensures that the GPU or Neural Processing Unit (NPU) pipelines remain fully saturated. This allows FALO to achieve state-of-the-art detection accuracy with exceptionally fast inference speeds on hardware that would otherwise choke on sparse convolutions.²⁷

For environments with extreme power and form-factor constraints, such as embedded systems directly within the autonomous vehicle or lightweight drones, the LiFT architecture (2025) targets real-time inference specifically on FPGA (Field Programmable Gate Array) platforms.²⁹ FPGAs are highly efficient but lack the massive memory bandwidth of GPUs. LiFT operates under aggressive, telecom-grade hardware constraints: a maximum computational limit of 30 GMACs (billion multiply-accumulate operations) and mandatory INT8 quantization for both weights and activations.²⁹ To achieve high precision under these routing and logic block constraints, LiFT discards traditional 3D voxels entirely in favor of 2D cell-based processing. This minimizes the need for skip connections that consume valuable FPGA routing resources and block RAM.²⁹ Its core mechanism, the Dual-bound Pillar Feature Net, boosts feature extraction dynamically without expanding computational complexity, enabling a total execution cost of just 20.73 GMACs while maintaining top-tier Mean Average Precision (mAP) on the NuScenes benchmark.²⁹

Multi-Modal Scheduling and Session Fusion

Modern perception systems rarely rely on LiDAR alone; they must fuse 3D point clouds with 2D camera imagery to provide rich semantic context and redundancy.⁴³ However, fusing separate 2D and 3D feature extraction backbones often leads to severe gradient conflicts during training and suboptimal scheduling during runtime due to differing execution latencies between the pixel pipelines and the point cloud pipelines.⁴⁶

The Attention-Enabled Point Fusion (AEPF) architecture mitigates this latency mismatch by introducing lightweight attention mechanisms to synchronize feature fusion.⁴⁶ Instead of complex, dense cross-attention matrices that scale quadratically and exhaust memory, AEPF uses a streamlined attention module to map semantic data from the image plane directly onto the voxelized point cloud space. This ensures that the scheduling of the two parallel hardware pipelines remains tightly coupled, preventing one modality from stalling while waiting for the other.⁴⁶

Viability for Edge and vRAN Environments

Autonomous perception systems operating over 5G/6G vRAN environments are the definitive "latency-critical" workloads targeted by CORA.⁴ In V2I offloading scenarios, the vehicle transmits raw or compressed LiDAR data to the edge RSU for processing. The data transformations introduced by FALO (converting sparse 3D data to 1D sequences) directly and profoundly influence network transmission strategies.²⁷ A linear 1D sequence is significantly easier to compress, packetize, and transmit sequentially over the cellular uplink than an unstructured, multi-dimensional 3D spatial map.⁹

Furthermore, integrating LiFT's ultra-lightweight execution profile (20.73 GMACs) directly onto telecom edge hardware ensures that the compute stage latency is absolutely minimized.²⁹ When CORA calculates the end-to-end latency budget for a V2I perception task, the deterministic and highly predictable execution times of FALO and LiFT allow the CORA coordinator to allocate a significantly larger portion of the tight latency budget to the unpredictable radio access layers.² This buffering drastically increases the overall reliability and safety margins of the autonomous system operating over the cellular network.

Heterogeneous Serving and Power-Aware Orchestration

In practical, large-scale deployments, edge computing nodes do not serve a single, isolated type of workload. They are fundamentally multi-tenant infrastructures, hosting a highly heterogeneous mix of NLP LLMs, continuous video analytics, and bursty spatial perception tasks simultaneously.⁴⁷ The inference systems managing the edge node must orchestrate these highly divergent workloads to maximize overall throughput while strictly adhering to environmental, thermal, and service-level constraints.³⁰

SLA-Driven Batching and Request Scheduling

The SLA-Driven ML Inference Framework, detailed at MLSys 2022, provides a highly robust architectural abstraction specifically designed for clouds and edge environments equipped with heterogeneous accelerators (e.g., mixing NVIDIA GPUs, Google TPUs, and specialized NPUs).³⁰ Traditional cluster schedulers treat ML inference as monolithic jobs, leading to severe inefficiencies. This SLA-driven framework breaks the inference pipeline down, generating heuristic, per-stage SLAs (Service Level Agreements) that map perfectly to the stage-based budgeting approach used by CORA.⁴

The framework utilizes a dynamic auto-scaler and an interference-aware GPU scheduler. It deeply profiles the application metrics in real-time through queue-proxy sidecar containers injected into the deployment (e.g., via Knative serving).³⁰ By explicitly defining the workload parameters—such as target latency bounds and hardware architecture requirements—the scheduler dynamically routes incoming requests to the optimal accelerator.³⁰ It pairs compute-heavy LLM prefill stages with high-tier GPUs, while simultaneously routing lightweight video frame filtering or MobileNet tasks to lower-tier accelerators.³⁰ This multi-dimensional, greedy heuristic optimization allows the system to survive highly bursty and fluctuating traffic patterns while maintaining the calculated end-to-end SLA for all tenants.

Hardware State and Power Management Mechanisms

As telecommunication operators aggressively scale their edge infrastructure, power consumption and thermal dissipation at the cell tower base become severe, uncompromising physical limiting factors.⁴ The μ -Serve framework directly addresses the critical gap of power-aware deep learning model serving in these environments.⁴ μ -Serve introduces an advanced mechanism for dynamic GPU frequency scaling that is mathematically linked directly to the application's SLOs.⁴

Traditional GPU drivers aggressively run the hardware at maximum clock speeds during execution to minimize latency, resulting in massive, inefficient power spikes. μ -Serve continuously monitors the "latency slack"—the difference between the task's actual execution time and its mandated deadline.⁴ If the slack is substantial (i.e., the task will easily finish before the deadline), μ -Serve safely steps down the GPU's operating frequency.⁴ Because hardware voltage scales non-linearly with frequency, this mechanism significantly reduces the total energy required per inference request. Comprehensive evaluations on production workloads reveal that μ -Serve achieves 1.2× to 2.6× power savings (representing up to a 61% absolute reduction in power consumption) without violating any SLO attainment targets.⁴

Viability for Edge and vRAN Environments

The integration of SLA-driven scheduling and power-aware frequency scaling is an absolute prerequisite for scalable vRAN and edge deployments. Cell towers have strictly capped, unyielding power supplies; exceeding these limits can trip breakers, overheat sealed telecom

cabinets, and cause catastrophic cascading failures across the RAN.

By employing μ -Serve⁴, edge operators can highly dense their computing deployments—installing more accelerators per physical cabinet—confident that the software layer will intelligently throttle power consumption based on real-time task urgency. When synchronized with the CORA framework, μ -Serve can utilize network conditions to inform its frequency scaling.² If CORA detects severe uplink congestion and calculates a delay in the arrival of inference data, μ -Serve can proactively lower the GPU frequency, saving power during the idle waiting period, and ramp it back up precisely when the packetized data arrives in memory.² This creates a holistic, power-aware cyber-physical system.

Cross-Domain Architectural Synthesis

Analyzing the trajectory of SOTA inference architectures from 2022 to 2026 reveals profound, cross-domain architectural convergences that are fundamentally reshaping how compute is deployed at the network edge.

The Abstraction of Memory as a Fluid Utility

Across all evaluated workloads, the most distinct and impactful trend is the transition away from static, application-managed memory toward unified, dynamic memory paging and state offloading. In LLM inference, PagedAttention¹⁵ proved that applying traditional OS-level virtual memory concepts to GPU VRAM eliminates the catastrophic fragmentation that plagues autoregressive generation. WEAVER takes this abstraction a step further, demonstrating that memory is not strictly bound to the execution state of a specific model; rather, attention state can be offloaded, written to shared memory, and processed by parallel instances (cold models) to maximize the global utility of the GPU's memory bandwidth.²¹

This paradigm is directly mirrored in video analytics. The Dynamic SinkCache mechanism²⁵ abandons the archaic concept of retaining all frames or a simple sliding window. It opts instead for a fluid memory state that retains only semantically relevant spatial-temporal tokens, dynamically evicting noise. By abstracting memory management away from strict, static allocations, edge nodes can seamlessly multiplex massive, complex models within the strict physical boundaries of telecom hardware.

Granularity of Scheduling: From Requests to Micro-Tasks

Historically, systems scheduled workloads at the macro-granularity of a "request" (e.g., process one image, generate one document). The state-of-the-art architectures have shattered this paradigm, shifting scheduling to a highly granular micro-task level. Sarathi-Serve fundamentally disrupts LLM scheduling by moving to iteration-level scheduling, dynamically composing heterogeneous execution blocks (prefill chunks mixed with decode tokens) to saturate hardware units evenly, rather than letting a single request monopolize the pipeline.¹⁶

Similarly, PacketGame dismantles the monolithic video stream, scheduling at the packet level

before the data even touches the hardware decoder.¹¹ By integrating combinatorial optimizers and temporal estimators at the ingress point, the system preempts wasted computation entirely. In the context of 5G vRAN, this micro-task scheduling perfectly mirrors the resource allocation mechanisms of the cellular MAC layer, which assigns Transmission Time Intervals (TTIs) and PRBs on a millisecond basis. The precise alignment of compute scheduling granularity (iteration/packet) with network scheduling granularity (PRB/TTI) is the foundational requirement that enables the cyber-physical synergy proposed by CORA.²

Restructuring Data for Hardware Parallelism

The uncompromising physical limitations of Edge and vRAN hardware—specifically regarding memory bandwidth and cache sizes—force algorithms to physically adapt their data structures. The FALO architecture's transformation of 3D sparse voxels into 1D linear sequences⁹ is a direct, engineered response to the reality that edge GPUs and NPUs cannot efficiently process irregular memory accesses. By forcing the data to conform to the hardware's optimal consumption pattern—contiguous linear reads—FALO avoids the massive latency penalties of cache misses.²⁷

This overarching design philosophy is echoed in LiFT's use of 2D cell processing to respect the strict routing limits of edge FPGAs²⁹, and in WEAVER's use of GPU-driven control flow to bypass the PCIe bottleneck.²¹ The definitive trend dictates that state-of-the-art inference algorithms can no longer treat hardware as an abstracted black box; they must be intimately co-designed with the specific parallelization constraints, memory hierarchies, and interconnects of edge silicon.

Strategic Implications for Future vRAN Deployments

The pursuit of latency-critical inference serving over cellular networks is governed by the immutable constraints of radio frequency transmission, edge compute capacity, and strict application SLOs. The CORA framework establishes conclusively that achieving reliable performance requires continuous, end-to-end coordination that intelligently partitions latency budgets across the uplink, computation, and downlink stages based on specific, profiled task characteristics.²

An exhaustive analysis of state-of-the-art inference architectures published between 2022 and 2026 reveals that the systems designed to operate within these compute stages have undergone radical, paradigm-shifting optimizations. For Large Language Models, architectures like vLLM, Sarathi-Serve, and WEAVER have successfully conquered memory fragmentation, head-of-line blocking, and multi-tenant scaling issues through PagedAttention, iteration-level scheduling, and GPU-driven dynamic control flow.¹⁵ In the domain of video analytics, systems like PacketGame demonstrate that pre-decode packet gating driven by contextual multi-view neural networks can reduce computational overhead by nearly 80%, preserving vital resources for downstream AI processing.¹¹ Meanwhile, 3D LiDAR perception frameworks like FALO and LiFT have proven that by aggressively restructuring data access patterns—transforming

complex 3D maps into 1D sequences or 2D cells—autonomous driving models can execute with ultra-low latency on severely constrained edge devices and FPGAs.²⁷

To ensure the continued viability and economic scaling of edge-assisted inference in expanding 5G and 6G ecosystems, the future of systems engineering lies in the deep, programmatic integration of these application-layer architectural mechanisms with the underlying radio network. When the context-aware packet gating of PacketGame, the iteration-level predictability of Sarathi-Serve, and the power-aware frequency scaling of μ -Serve are fully synthesized with the dynamic, stage-based latency routing of CORA, the cellular edge will possess the resilience, efficiency, and scale required to power the next generation of pervasive artificial intelligence.

Works cited

1. End-to-End Coordination of RAN and Edge Server for Latency, accessed on April 16, 2026, https://experts.colorado.edu/display/pubid_504668
2. 논문 | 이경한 교수 연구실 | 서울대학교 전기·정보공학부 | 디씨클, accessed on April 16, 2026, <https://app.rndcircle.io/lab/8530d718-9acc-416d-a9ef-cf0c76c9456f/papers>
3. Sunghyun Jin's research works | Seoul National University and other places - ResearchGate, accessed on April 16, 2026, <https://www.researchgate.net/scientific-contributions/Sunghyun-Jin-2236911278>
4. End-to-End Coordination of RAN and Edge Server for Latency-Critical Inference Serving over Cellular Networks | Request PDF - ResearchGate, accessed on April 16, 2026, https://www.researchgate.net/publication/397970817_End-to-End_Coordination_of_RAN_and_Edge_Server_for_Latency-Critical_Inference_Serving_over_Cellular_Networks
5. Online Scheduling of Moldable Task Graphs under Common Speedup Models | Request PDF - ResearchGate, accessed on April 16, 2026, https://www.researchgate.net/publication/367158732_Online_Scheduling_of_Moldable_Task_Graphs_under_Common_Speedup_Models
6. Visual Modality Prompt for Adapting Vision-Language Object Detectors - arXiv, accessed on April 16, 2026, <https://arxiv.org/html/2412.00622v1>
7. Adaptation of Deep Object Detectors for New Modalities - Espace ETS, accessed on April 16, 2026, https://espace.etsmtl.ca/id/eprint/3755/1/Rapela%20Medeiros_Heitor.pdf
8. Open World Object Detection: A Survey - arXiv, accessed on April 16, 2026, <https://arxiv.org/html/2410.11301v1>
9. Backbone architecture of YOLOv6. | Download Scientific Diagram - ResearchGate, accessed on April 16, 2026, https://www.researchgate.net/figure/Backbone-architecture-of-YOLOv6_fig3_392166846
10. Applications of artificial intelligence in battling against covid-19: A literature review - PMC, accessed on April 16, 2026,

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC7532790/>
11. PacketGame: Multi-Stream Packet Gating for Concurrent ... - Mu Yuan, accessed on April 16, 2026, https://yuanmu97.github.io/preprint/packetgame_sigcomm23.pdf
 12. SplitStream: Distributed and workload-adaptive video analytics at the edge - Temple CIS, accessed on April 16, 2026, https://cis.temple.edu/~wu/research/publications/Publication_files/splitstream-jnc a.pdf
 13. Tutti: coupling 5G RAN and mobile edge computing for latency-critical video analytics | Request PDF - ResearchGate, accessed on April 16, 2026, https://www.researchgate.net/publication/364527845_Tutti_coupling_5G_RAN_and_mobile_edge_computing_for_latency-critical_video_analytics
 14. The 2025 Conference on Empirical Methods in Natural Language Processing, accessed on April 16, 2026, <https://aclanthology.org/events/emnlp-2025/>
 15. Power-aware Deep Learning Model Serving with μ -Serve - ResearchGate, accessed on April 16, 2026, https://www.researchgate.net/publication/381317142_Power-aware_Deep_Learnin g_Model_Serving_with_-Serve
 16. Foundations of Large Language Models - Mick McQuaid, accessed on April 16, 2026, <https://mickmcquaid.com/promptEngr/Xiao2025.pdf>
 17. chenhongyu2048/LLM-inference-optimization-paper - GitHub, accessed on April 16, 2026, <https://github.com/chenhongyu2048/LLM-inference-optimization-paper>
 18. Attention Optimization - Aussie AI, accessed on April 16, 2026, <https://www.aussieai.com/research/attention>
 19. LegoDiffusion: Micro-Serving Text-to-Image Diffusion Workflows - arXiv, accessed on April 16, 2026, <https://arxiv.org/html/2604.08123v1>
 20. ALADDIN: JOINT PLACEMENT AND SCALING FOR SLO-AWARE LLM SERVING - OpenReview, accessed on April 16, 2026, <https://openreview.net/pdf?id=9Gmw9LuApj>
 21. ATC 2025 | Awesome Papers, accessed on April 16, 2026, <https://paper.lingyunyang.com/reading-notes/conference/atc-2025>
 22. Weaver: Efficient Multi-LLM Serving with Attention Offloading | USENIX, accessed on April 16, 2026, <https://www.usenix.org/conference/atc25/presentation/gao>
 23. ACM SIGCOMM 2023 Program - Events, accessed on April 16, 2026, <https://conferences.sigcomm.org/sigcomm/2023/program.html>
 24. Aha - Predicting What Matters Next: Online Highlight Detection Without Looking Ahead - arXiv, accessed on April 16, 2026, <https://arxiv.org/html/2509.16421v2>
 25. AHA -- Predicting What Matters Next: Online Highlight Detection Without Looking Ahead - arXiv, accessed on April 16, 2026, <https://arxiv.org/pdf/2509.16421>
 26. Aha! - Predicting What Matters Next: Online Highlight Detection Without Looking Ahead | Read Paper on Bytez, accessed on April 16, 2026, https://bytez.com/docs/neurips/119707/paper?_c=eyJ2ljoxLCJyZWxhdGVkljpbImN vZGUlLCJyZWZlcmVuY2VzliwiY29uZmVyZW5jZSJdfQ%3D%3D
 27. FALO: Fast and Accurate LiDAR 3D Object Detection on Resource-Constrained Devices, accessed on April 16, 2026, <https://arxiv.org/html/2506.04499v1>

28. FALO: Fast and Accurate LiDAR 3D Object Detection on Resource-Constrained Devices, accessed on April 16, 2026, https://www.researchgate.net/publication/392466242_FALO_Fast_and_Accurate_LiDAR_3D_Object_Detection_on_Resource-Constrained_Devices
29. LiFT: Lightweight, FPGA-tailored 3D object detection based on LiDAR data - arXiv, accessed on April 16, 2026, <https://arxiv.org/html/2501.11159v1>
30. SLA-Driven ML INFERENCE FRAMEWORK FOR CLOUDS WITH HETEROGENEOUS ACCELERATORS - MLSys Proceedings, accessed on April 16, 2026, https://proceedings.mlsys.org/paper_files/paper/2022/file/bcf9bef61a534d0ce4a3c55f09dfcc29-Paper.pdf
31. [Solution] IPA: Inference Pipeline Adaptation to achieve high accuracy and cost-efficiency - NSF Public Access Repository, accessed on April 16, 2026, <https://par.nsf.gov/servlets/purl/10443141>
32. A House United Within Itself: SLO-Awareness for On-Premises Containerized ML Inference Clusters via Faro - arXiv, accessed on April 16, 2026, <https://arxiv.org/html/2409.19488v1>
33. Towards Efficient Generative Large Language Model Serving: A Survey from Algorithms to Systems - arXiv, accessed on April 16, 2026, <https://arxiv.org/html/2312.15234v1>
34. 面向大语言模型训练的智能计算系统综述 - JCST, accessed on April 16, 2026, <https://jcst.ict.ac.cn/cn/article/doi/10.1007/s11390-024-4178-1>
35. Pangu-Agent: A Fine-Tunable Generalist Agent with Structured Reasoning - arXiv, accessed on April 16, 2026, <https://arxiv.org/pdf/2312.14878>
36. Chronos: Prescheduled circuit switching for LLM training - McKeown Group, accessed on April 16, 2026, <https://yuba.stanford.edu/~nickm/papers/chronos2025.pdf>
37. Dynamic Hierarchical Birkhoff-von Neumann Decomposition for All-to-All GPU Communication - arXiv, accessed on April 16, 2026, <https://arxiv.org/pdf/2602.22756>
38. LEVERAGING APPLICATION KNOWLEDGE FOR HIGH-PERFORMANCE NETWORKS: FROM PRESCHEDULED CIRCUIT SWITCHING TO RUNTIME VERIFICATION A DI - McKeown Group, accessed on April 16, 2026, <https://yuba.stanford.edu/~nickm/papers/sundararajan-renganathan-thesis.pdf>
39. ACM SIGCOMM 2023 List of Accepted Papers - Events, accessed on April 16, 2026, <https://conferences.sigcomm.org/sigcomm/2023/list-accepted.html>
40. EdgeEye: An Edge Service Framework for Real-time Intelligent Video Analytics, accessed on April 16, 2026, <https://www.semanticscholar.org/paper/EdgeEye%3A-An-Edge-Service-Framework-for-Real-time-Liu-Qi/a872ca8a5f87fd2a105cee47bba80366ef1fc2ab>
41. EdgeEye: An Edge Service Framework for Real-time Intelligent Video Analytics | Request PDF - ResearchGate, accessed on April 16, 2026, https://www.researchgate.net/publication/325436389_EdgeEye_An_Edge_Service_Framework_for_Real-time_Intelligent_Video_Analytics
42. Internet-of-Things Edge Computing Systems for Streaming Video Analytics: Trails Behind and the Paths Ahead - MDPI, accessed on April 16, 2026,

- <https://www.mdpi.com/2624-831X/4/4/21>
43. Deep learning for LiDAR-only and LiDAR-fusion 3D perception: a survey, accessed on April 16, 2026, <https://www.oaepublish.com/articles/ir.2021.20>
 44. Opportunities, Applications, and Challenges of Edge-AI Enabled Video Analytics in Smart Cities - IEEE Xplore, accessed on April 16, 2026, <https://ieeexplore.ieee.org/iel7/6287639/6514899/10198424.pdf>
 45. Computer Vision and Pattern Recognition Jun 2025 - arXiv, accessed on April 16, 2026, <https://www.arxiv.org/list/cs.CV/2025-06?skip=50&show=2000>
 46. AEPF: Attention-Enabled Point Fusion for 3D Object Detection - PMC - NIH, accessed on April 16, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11398036/>
 47. USENIX ATC '25 Technical Sessions, accessed on April 16, 2026, <https://www.usenix.org/conference/atc25/technical-sessions>
 48. Systems Research Group | Asia - Microsoft Research: Publications, accessed on April 16, 2026, <https://www.microsoft.com/en-us/research/group/systems-research-group-asia/publications/>
 49. A House United Within Itself: SLO-Awareness for On-Premises Containerized ML Inference Clusters via Faro - Distributed Protocols Research Group (DPRG) - University of Illinois, accessed on April 16, 2026, https://dprg.cs.uiuc.edu/data/files/2025/Faro_Eurosys2025.pdf